ABISHEK PANDIAN S

Data Science & ML Engineer • Computer Vision • Python Developer

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PROFESSIONAL SUMMARY

Results-driven Data Science fresher and MCA candidate with proven, hands-on experience building two production-grade ML systems: a deep learning reproductive cancer detection platform (MobileNetV2 / TensorFlow) and a real-time computer vision fire detection pipeline (OpenCV). Skilled in end-to-end ML delivery — from exploratory data analysis and model training to Flask web deployment and automated PDF reporting. Seeking to contribute to high-impact AI/ML teams locally or internationally.

TECHNICAL SKILLS

AI / ML: Machine Learning, Deep Learning, CNN, Transfer Learning, MobileNetV2, TensorFlow, Keras, Scikit-learn

Computer Vision: OpenCV (cv2), Pillow (PIL), Image Processing, Feature Extraction

Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn, EDA, Precision / Recall / F1-Score

Web & Deployment: Flask, Flask-Mail, SMTP, ReportLab (PDF), HTML, CSS

Dev Tools: Python 3.8+, SQL, PHP | Jupyter Notebook, Google Colab, VS Code, Git, GitHub

WORK EXPERIENCE

Data Science Intern – **Sharpened Mind Tech & Solutions** Madurai, India (On-site) *Sep 2025 – Oct 2025*

Project: Automated Fire Detection System for Chemical Industry | Python · OpenCV · NumPy · Pandas · Matplotlib · SMTP · GitHub | GitHub Repo

- Achieved **~92% fire detection accuracy** on live video streams, measured by **precision, recall & F1-score**, by building a real-time OpenCV + NumPy computer vision pipeline with deep learning-inspired colour and shape feature extraction.
- Reduced false-alarm rate by **~40%** and cut incident-to-response time to **under 3 seconds** by integrating automated model evaluation triggers with SMTP email alerts and hardware valve shut-off controls in Python.
- Delivered a production-ready, documented pipeline on schedule by performing **EDA** on 500+ video frames using Pandas and Matplotlib to validate data quality before model training, and managing all code via **GitHub** version control.

PROJECTS

Anonymous Detection & Staging of Reproductive Cancer – **University Capstone Project | MobileNetV2 + Flask Web App** —GitHub

Stack: Python · TensorFlow · Keras · MobileNetV2 · Scikit-learn · OpenCV · Pillow · NumPy · Pandas · Matplotlib · Seaborn · Flask · ReportLab · Flask-Mail/SMTP · HTML · CSS | VS Code · Jupyter Notebook · Google Colab

- Designed and trained a **MobileNetV2 transfer learning** model for high-accuracy reproductive cancer image classification and multi-stage detection, using TensorFlow/Keras with Scikit-learn evaluation metrics (**accuracy, precision, recall, F1**).
- Built a full-stack **Flask** web application enabling anonymous patient submissions; automated personalised PDF medical reports via **ReportLab** and email delivery via **Flask-Mail/SMTP** for seamless treatment planning workflow.
- Conducted in-depth **EDA** on medical imaging datasets using **Pandas**, **Matplotlib**, and **Seaborn** to ensure data quality and class balance before model training, improving generalisation performance.

Supermarket Management System – **Personal Project**

Built a full-stack billing and customer management application with invoicing, coupon generation, and persistent records using **Python**, **PHP**, **MySQL**, HTML, and CSS — demonstrating full-stack and database design skills.

AI-Powered HR Policy Chatbot – **Personal Project | Groq API (LLaMA 3.1) + Streamlit Web App** —GitHub

Stack: Python · Groq API · LLaMA 3.1 · Streamlit · NLP · Git · GitHub

- Built an intelligent HR Policy Q&A chatbot by integrating **Groq API** with **LLaMA 3.1** (8B), enabling employees to query leave, WFH, salary, and onboarding policies in natural language — delivering accurate, context-aware responses in real time.
- Designed a modular **RAG-style architecture** — keyword-based policy retrieval engine (**policy_data.py**) feeding structured context into the LLM prompt — reducing hallucinations and ensuring policy-grounded, professional answers.
- Deployed a full **Streamlit** web application with persistent multi-turn **chat history**, session state management, and a clean chat UI — demonstrating end-to-end GenAI application development and cloud deployment skills.

AI Security Gate — Automated Secret Detection Pipeline – **Personal Project | GitHub Actions + TruffleHog AI Scanner** —GitHub

Stack: GitHub Actions · TruffleHog AI · CI/CD · YAML · DevSecOps · Git

- Built an automated **CI/CD security gate** using **GitHub Actions** that triggers on every push to automatically block accidental exposure of API keys, passwords, and credentials — preventing security breaches before code reaches production.
- Integrated **TruffleHog AI** — an AI-driven secret scanner — to perform deep **verified secret detection** across the full Git commit history, automatically failing the build when real leaked credentials (AWS keys, passwords) are found.
- Demonstrated **DevSecOps** best practices by embedding security scanning directly into the development workflow — showcasing knowledge of **secure coding, CI/CD pipelines**, and AI-assisted vulnerability detection.

EDUCATION

Master of Computer Applications (MCA) 6Madurai Kamaraj University, Madurai, India *2024 – 2026*

Bachelor of Science in Computer Science (BSc CS) N.M.S.S. Vellaichamy Nadar College, Madurai, India *2021 – 2024 | CGPA: 7.3 / 10*

ACHIEVEMENTS & LANGUAGES

2nd Prize — Debate Competition, Power Grid Corporation of India (Vigilance Awareness Board, theme: "How to Reduce Bribery")

Languages: Tamil (Native) | English (Professional Working Proficiency)